

Citrus Disease Detection Using Image Processing

Saif Mohamed, Hassan Abdallah, Marawan Mohamed, Abdelrahman Medhat & Mohamed Fathi

Abstract

Various problems associated with such plant diseases as Citrus Canker(CC) or Black Spot result in numerous economic losses within the field of agriculture. Therefore, it becomes critically important to develop an algorithm that can detect the presence of a certain disease by carrying out binary classification of the input image in an accurate way. The paper proposes an application of transfer learning in order to achieve the optimal optimization of a ResNet50 model. Thus, this method can help extract useful features of the input images through deep residual connections that mitigate the vanishing gradient problem. In order to test the efficiency of the model suggested, a comparative analysis was conducted between the model and other Deep Learning(DL) models such as MobileNetV2, VGG16, and EfficientNetB0. An appropriate dataset, which is used for the purpose of training, was provided. It should be mentioned that the image augmentation process was applied during the training stage of the model, which positively affected its efficiency. Thus, it could be stated that the ResNet50 model managed to surpass others in terms of the accuracy score equal to 97.79% along with the precision, recall, and F1-score of 0.9779 as well. VGG16 reached an accuracy of 97.54%, EfficientNetB0 reached 97.05%, and MobileNetV2 reached 95.33%.

1. Introduction

Apart from their vital role in promoting human health, citrus fruits are extremely important from the nutritional and economic standpoint. Regarding nutrition, citrus fruits contain high amounts of minerals and vitamins; specifically, they contain vitamin C that helps to increase immunity and prevent a wide range of diseases because of its strong antioxidative effect [\[1\]](#). On top of that, along with nutrition, antioxidative, anti-inflammatory, and immunomodulatory qualities of citrus juices can protect human body from any potential pathogen attacks [\[2\]](#).

Additionally, this characteristic applies not only to the fruits themselves but also to citrus peels since active compounds in these fruits' peels also demonstrate strong antioxidative activities [\[3\]](#). All in all, mentioned above features show how important are citrus fruits for people.

Such high importance of citrus fruits for people contributes to their high economic importance as they are widely grown and distributed worldwide and become economically and socially significant

crops [\[4\]](#). To prove this statement, one might say that citrus fruits are the most exported fresh produce in the world by volume [\[5\]](#).

CC is an extremely hazardous bacterial infection that continuously endangers the production of citrus fruits worldwide [\[6\]](#). Specifically, CC is a bacterial infection caused by an extremely aggressive bacterial species known as *Xanthomonas citri* subsp. *citri* (Xcc), which is one of the most severe infections that negatively impact the production of important citrus fruit crops cultivated throughout the world [\[7\]](#). Furthermore, the identification and analysis of the genes of *Citrus sinensis* in regard to their plant-bacteria interactions are crucial in managing several citrus pathogens [\[8\]](#).

Citrus crops are of immense commercial importance globally, yet their growth faces numerous challenges, such as being highly susceptible to a range of plant diseases, including CC and Citrus Black Spot (blemishes), that reduce yield and quality [\[9\]](#). The conventional process of visual disease detection through the eyes of experts is slow, lacks accuracy, is costly, and requires a high level of intervention [\[10\]](#). There is, therefore, an urgent need to employ image processing and transfer learning methodologies to automate such operations.

Over the years, DL techniques have developed significantly, and researchers can now use them to detect and classify objects within images with exceptional precision. Indeed, several studies have used advanced DL algorithms to evaluate and implement agricultural processes with better results than classical methods [\[11\]](#). For instance, the employment of deep Convolutional Neural Networks(CNN) has facilitated the detection and classification of multiple citrus leaf and fruit diseases, thus improving crop management [\[12\]](#).

Despite the aforementioned developments, image-based plant disease recognition in crops is a relatively challenging problem, especially considering that traditional methods involved image recognition based on handcrafted feature extraction, which was labor-intensive, needed expertise, and did not always achieve optimal detection results [\[13\]](#).

While there is sufficient evidence that demonstrates the potential of DL models in detecting severe infections, such as CC [\[14\]](#), most machine vision solutions either demand vast computational power or cannot distinguish between multiple diseases. There exists, therefore, a vital research gap: the development of efficient and lightweight models to discriminate among confusing citrus diseases for edge deployment [\[15\]](#).

Therefore, this study aims to develop a CNN-based DL model capable of classifying orange fruit images into two categories: CC and Citrus Black Spot, to provide an accurate and automated solution for citrus disease detection and differentiation.

The prime objective of this research is to develop an automated detection and classification model for various diseases of citrus plants. In this research, emphasis has been placed upon the following two diseases:

1-CC

2-Citrus Black Spot

The objectives of this research include the following:

1-Improving the quality of the image data through the use of image pre-processing techniques including noise reduction, contrast enhancement, and normalization to prepare the data set for subsequent analysis.

2-Extraction of visual feature vectors from the disease images through recognition of characteristic morphological and textural features associated with CC and Citrus Black Spot.

3-Development of a robust model for classification of images as CC or Citrus Black Spot.

Some of the key goals of this study were to propose a highly efficient and accurate model for the automatic identification and classification of diseases affecting citrus crops. Hence, the major achievements made in the course of this study include:

The first contribution made was in terms of proposing a well-optimized algorithm for the automated detection of citrus diseases using the ResNet50 model. In particular, the developed model demonstrated strong performance in distinguishing between two of the most common citrus diseases – CC and Citrus Black Spot, with 97.79% accuracy.

Secondly, in order to demonstrate the superiority of the proposed algorithm, a comparative study was carried out with the aid of three other advanced deep learning networks, namely MobileNetV2, VGG16, and EfficientNetB0. The comparison was done based on equal experimental conditions, and as expected, ResNet50 was shown to offer the best classification accuracy among the four models used in the study.

Another achievement of the study included the optimization of the training process through the integration of transfer learning, dynamic learning rates, and extensive data augmentation techniques.

Lastly, all models used in the study were subjected to various evaluation criteria such as Precision, Recall, F1-Score, and Confusion Matrix.

2. Related Work

Recent research in citrus disease detection has moved toward systems that are not only accurate in experiments but also useful in real agricultural environments. The study in [\[9\]](#) showed that combining convolutional neural networks with hyperspectral imaging can improve citrus disease classification because spectral data provide information that ordinary RGB images may miss.

Efficiency has also become a central requirement. Reference [\[16\]](#) introduced a lightweight citrus leaf disease detection model designed for complex orchard scenes, where uneven lighting, small disease spots, and background interference can reduce model performance. This direction is important because a practical system should work on low-cost devices and support fast diagnosis in the field.

Hybrid learning methods have also been explored to improve reliability. The work in [\[17\]](#) combined DL architectures with machine learning classifiers for citrus disease classification and reported strong performance for diseases such as black spot, CC , and greening. Together, these studies show that the field is shifting from accuracy-only models toward solutions that balance accuracy, speed, robustness, and practical deployment.

2.1. Traditional Image Processing Methods

Before DL became dominant, plant disease detection mainly depended on image processing and handcrafted features. Texture and statistical feature analysis remain important because disease lesions often appear as rough, spotted, or irregular surface patterns. The Gray-Level Co-occurrence Matrix (GLCM)-based dataset analysis in [\[18\]](#) showed that texture measures can support plant disease recognition and can also help evaluate whether a dataset is reliable for model training.

A related comparative study in [\[19\]](#) combined GLCM and Local Binary Pattern (LBP) features with an Support Vector Machine (SVM) classifier, providing an interpretable and computationally efficient alternative to fully deep learning-based systems.

Although these methods are simple and efficient, they are sensitive to lighting variation, image noise, shadows, and complex backgrounds. For citrus diseases such as canker and black spot, traditional methods may fail when lesions are small, visually similar, or captured under natural orchard conditions.

2.2. DL Approaches

DL has significantly improved plant disease detection because it learns useful visual features directly from raw images instead of depending only on handcrafted descriptors. Modern models can recognize lesion shape, color distribution, texture, and surrounding context more effectively than classical methods.

Lightweight detection models are especially important for real-time agricultural use. The PEW-YOLO model proposed in [20] improved citrus pest and disease detection while reducing model parameters, making it more suitable for field and edge-device applications.

Recent YOLO-based methods have also focused on improving detection under complex natural conditions. The YOLO-Citrus model in [13] was designed for citrus leaf disease detection in complex agricultural environments and reported strong detection performance with reduced model size, while the ROI-YOLOv10 approach in [21] improved detection by focusing the model on disease-relevant regions before classification.

Classification-based DL has also achieved strong results. EfficientNet-B5 was applied in [22] to classify citrus diseases including black spot, canker, greening, and healthy samples. In addition, the hybrid meta-heuristic deep learning framework in [23] used feature selection and transfer learning to improve classification performance on citrus leaf and fruit disease datasets.

Despite these improvements, DL models still face challenges related to dataset quality, computational cost, and real-world generalization. A model that performs well on a prepared benchmark dataset may not always perform equally well in orchards where lighting, background, fruit orientation, and disease symptoms vary.

2.3. Research Gap

Although recent studies have improved citrus disease detection, several gaps remain. Traditional image processing methods are computationally simple, but they are not robust enough for complex field images. DL models provide better accuracy, but many of them require large datasets, high computing resources, and carefully controlled training conditions.

A major limitation in the literature is the limited attention given to fine-grained classification between visually similar citrus diseases. CC and black spot can both appear as dark or irregular lesions on fruit and leaf surfaces, which increases the risk of misclassification when the model does not learn enough discriminative features.

The field-based citrus classification study in [24] highlights this challenge clearly because its orange dataset focuses specifically on black spot and CC captured in open orchard settings with variations in lighting, background, and fruit orientation. This confirms that real-world citrus disease classification is more difficult than controlled image classification.

Therefore, the present study focuses on developing a robust and efficient model that can distinguish between visually similar citrus diseases while remaining suitable for practical agricultural use. The goal is not only to achieve high accuracy but also to improve reliability under realistic conditions.

Table 1. List of Various Research Studies

Ref.	Techniques	Dataset	Accuracy	Key Contribution
[9]	CNN + Hyperspectral Imaging	Citrus fruits and leaves	92.1%	Combined spectral and spatial information to improve citrus disease classification.
[16]	YOLOv8n-CDDA with ARMS and attention	Citrus leaf disease dataset	90.89%	Built a lightweight detector for complex orchard backgrounds and edge deployment.
[17]	Hybrid DL + ML classifiers	Citrus disease images	98.32%	Combined CNN architectures with machine learning classifiers for robust classification.
[18]	GLCM metric analysis + DarkNet19	Plant disease image datasets	91%	Used texture metrics to evaluate dataset quality and support disease recognition.
[19]	GLCM + LBP + SVM	Plant leaf images	91.73%	Provided interpretable handcrafted texture features with low computational cost.
[20]	PEW-YOLO	Citrus leaf pest and disease dataset	88.6% mAP50	Improved lightweight detection and reduced model parameters for real-time use.
[13]	YOLO-Citrus	Citrus leaf disease field images	96.6% mAP@0.5	Designed a lightweight YOLOv11-based model for complex agricultural environments; also reported 81.6% mAP@0.5:0.95.
[21]	ROI + YOLOv10	Citrus disease detection dataset	98.4% mAP	Improved detection precision by focusing on disease-relevant image regions.
[22]	EfficientNet-B5 + SMOTE + augmentation	Kaggle citrus disease dataset	99.22%	Classified black spot, canker, greening, and healthy citrus samples with high accuracy.
[23]	DenseNet-201 + AlexNet + meta-heuristic optimization	Hybrid citrus leaf/fruit dataset	99.6%	Used feature selection and transfer learning to improve citrus disease classification.

Ref.	Techniques	Dataset	Accuracy	Key Contribution
[24]	CNN + ViT + SVM ensemble	Real-field orange and lemon citrus datasets	97% orange; 99% lemon	Addressed visually similar citrus diseases, especially black spot and CC, under field-style conditions.

3.Methodology

To be able to see how each component is related in the entire system, it will be important to first draw a diagram on how each operation works. This will be done to facilitate an understanding of the process before discussing each component. Our described pipeline is very simple and easy in terms of operation. This involves using an image of the citrus fruit as the input to our system. This image is then processed in order to improve its resolution and enhance it. Once enhanced, it is then fed into our network structure, whereby it becomes the object from which we extract features.

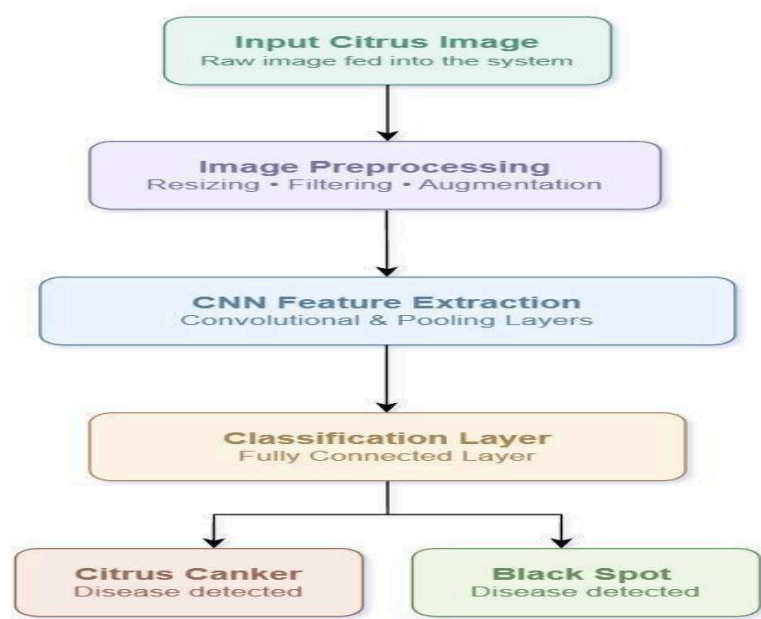


Figure 1: The proposed system architecture for citrus disease classification

3.1. Dataset Collection and Description

It is believed that the first step in resolving this issue will be choosing the appropriate data. The primary data used in this study was acquired from an open-source Kaggle repository. Utilizing Kaggle datasets has become a widely accepted and validated practice in recent citrus disease diagnostic frameworks [22]. For our specific methodology, the selected dataset consists of a total of 2,439 images of oranges. Considering that the final output is supposed to be a binary classifier,

the number of classes was limited to two major ones, which could affect the surface of the fruits under study. That is why we have focused only on CC and Black Spot. Rather than performing the split manually, the pre-existing directories were used. The training set consists of 2,032 images (Black Spot = 1,030, CC = 1,002), and the testing one includes 407 images (Black Spot = 206, CC = 201). Thus, this way of working allows for a standardized evaluation procedure when the classifier studies the images in the training set and is later tested on an entirely new set of data.

3.2. Preprocessing

However, the acquired raw images required essential preprocessing steps prior to model training. Firstly, all images were resized to 224 x 224 pixels. This method is beneficial since it increases the speed of learning and corresponds to the input image size in popular CNN architectures. Secondly, the problem of image noise was solved with the help of normalization and application of filters. As a result, diseased spots became more visible. Lastly, to prevent our algorithm from learning the images, we applied data augmentation. In particular, flipping, rotating, and zooming images helped to achieve diversity within the training dataset. In such a manner, we managed to teach our algorithm to recognize the actual disease and not just remember the training images.

3.3. Proposed Model

In this research, we introduce a deep CNN architecture that could help in achieving high classification accuracy rates in the process of classifying citrus disease. In this regard, ResNet50 is selected as the primary classifier in this study due to its highly accurate performance in plant disease detection, as well as its ability to train very deep networks effectively [\[25\]](#). Different from traditional CNNs which suffer from vanishing gradient problems as depth increases, the proposed ResNet50 architecture introduces residual (skip) connections that allow gradients to flow directly through the network. This design enables the model to learn rich hierarchical features across 50 layers without degradation in performance.

This proposed CNN architecture can implement transfer learning, and logically divide itself into two main parts. The first part is the base feature extractor. As mentioned above, the ResNet50 base model has been pre-trained with a huge database of images (called ImageNet). Therefore, this model can effectively extract the visual features, including texture, color, and boundaries of lesions, from the input citrus fruits.

Since the problem in this study involves the binary classification of citrus diseases, it is necessary to develop a new design of the classification head that should be stacked after the ResNet50 base for making the transfer learning possible. The architecture of the customized classification head is structured as follows:

Global Average Pooling 2D: Reduces the size of the feature map and hence saves computing resources.

Batch Normalization: The incorporation of Batch Normalization can lead to the stable convergence of the model by normalizing the input of each layer.

Fully Connected Dense layer (256 units, activation: ReLU): Dense layer allows the model to learn nonlinear functions of features.

Dropout (rate=0.4): The use of dropout helps avoid overfitting during the training by randomly dropout neurons with a dropout rate of 40%.

Output Layer: A fully connected dense layer with activation function as sigmoid and outputs a scalar value between 0 and 1 as the binary classification results.

Finally, in order to demonstrate the effectiveness of the proposed ResNet50 architecture, comparative experiments have been conducted against three other widely used pre-trained CNN structures: EfficientNetB0, VGG16, and MobileNetV2.

These architectures continue to be extensively utilized and validated as reliable baselines in recent agricultural DL studies [[26](#),[27](#),[28](#)].

3.4. Training Strategy

The selection of the proper constraints is equally important to that of the architecture of the model. The Adam optimizer [[29](#)] was selected for solving the optimization problem due to its recently proven superiority and high stability in modern image classification tasks compared to other standard optimizers. The Binary Crossentropy algorithm was chosen as the loss function, which is a typical choice because the number of classes equals two. To enhance the optimization process, we used a learning rate scheduler technique to ensure the best results during the whole training process. It enables the automatic reduction of the learning rate when there are no more improvements.

Besides, the checkpoint technique was applied to track the training process and save the best model weights.

4. Experiments and Results

The dataset consists of 2,439 images, split into 2,032 for training and 407 for testing. Prior to model training, all images were resized to 224 x 224 pixels. Data augmentation techniques, including flipping, rotating, and zooming, were applied to achieve diversity within the training dataset and prevent the algorithm from merely remembering the images.

The model optimization utilized the Adam optimizer along with a Binary Crossentropy loss function, a learning rate scheduler technique, and checkpoint techniques to track the training process and save the best model weights.

To rigorously evaluate the model's diagnostic performance, standard quantitative classification metrics were employed. They include essential indicators such as Accuracy, Precision, Recall, and F1-Score [30]. Accuracy provides an idea of the overall percentage of correct classifications. Precision reflects how many of those samples that are marked as sick actually are ill. Recall evaluates our ability to discover sick samples among the testing set. In order to get a compromise between two indicators, the F1-Score is calculated. Finally, a Confusion Matrix is created to understand whether the algorithm confused CC with Black Spot too often.

The mathematical equations used for these metrics are as follows:

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN}) \quad (1)$$

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP}) \quad (2)$$

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN}) \quad (3)$$

$$\text{F1-Score} = 2 * (\text{Precision} * \text{Recall}) / (\text{Precision} + \text{Recall}) \quad (4)$$

Where :

- *TP : True Positive.*
- *TN : True Negative.*
- *FP : False Positive.*
- *FN : False Negative*

The effectiveness of the proposed approach has been carefully evaluated using the generated models on 407 images, not included into training.

This test was conducted according to the standards for accuracy, precision, recall, and F1-score described in the methodology section. Comparative performance of the proposed framework (ResNet50) in relation to other base ones can be seen in Table 2.

Table 2. Comparative Performance of the Evaluated DL Models

Model Architecture	Precision	Recall	F1-Score	Accuracy(%)
MobileNetV2	0.9546	0.9533	0.9532	95.33%
EfficientNetB0	0.9706	0.9705	0.9705	97.05%
VGG16	0.9754	0.9754	0.9754	97.54%
ResNet50 (Proposed)	0.9779	0.9779	0.9779	97.79%

Based on the results presented in Table 2, it can be stated that the proposed ResNet50 model showed the best results compared to others by giving the highest accuracy rate of 97.79%. Furthermore, all of its metrics were balanced, since precision, recall, and F1-score rates were all equal to 0.9779, which reflects high robustness of the suggested model and its efficiency in recognizing CC and Black Spot. VGG16 followed closely with 97.54% accuracy, while EfficientNetB0 reached 97.05% and MobileNetV2 reached 95.33%. The strong performance of ResNet50 is attributed to its deep residual connections, which allow the model to learn rich hierarchical features without suffering from vanishing gradients during training.

Moreover, Confusion Matrices, generated during the experiments, confirm the strong performance of the suggested algorithm. Indeed, ResNet50 successfully classified 398 out of 407 images correctly, with only 5 Black Spot samples misclassified as Canker and 4 Canker samples misclassified as Black Spot. Therefore, the generated model can be considered highly effective in recognizing similar diseases.

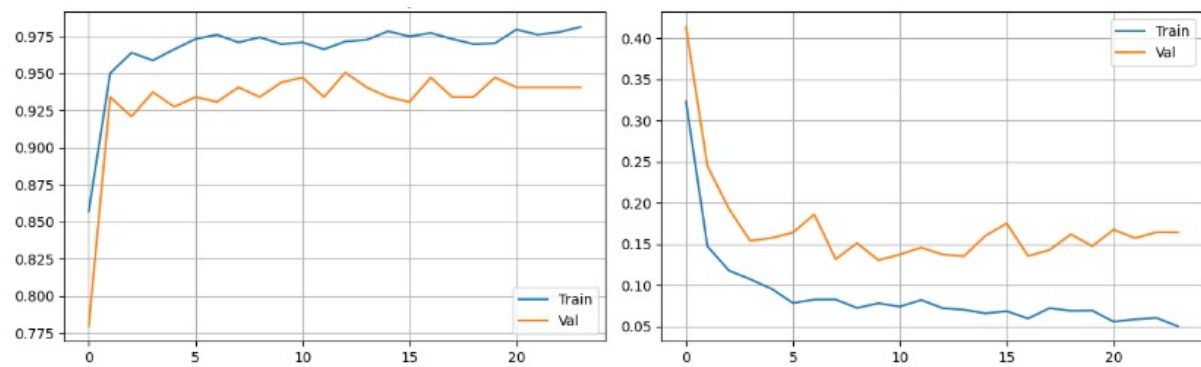


Figure 2: Accuracy & Loss Curves of ResNet50

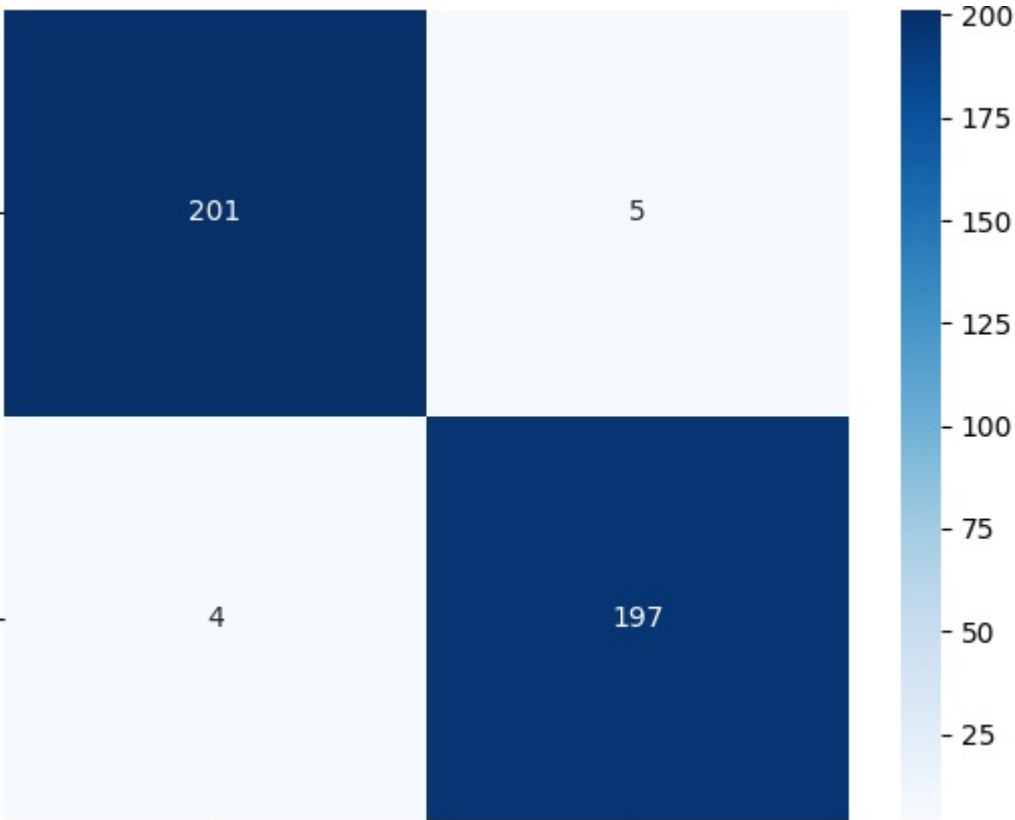


Figure 3: Validation Confusion Matrix of ResNet50

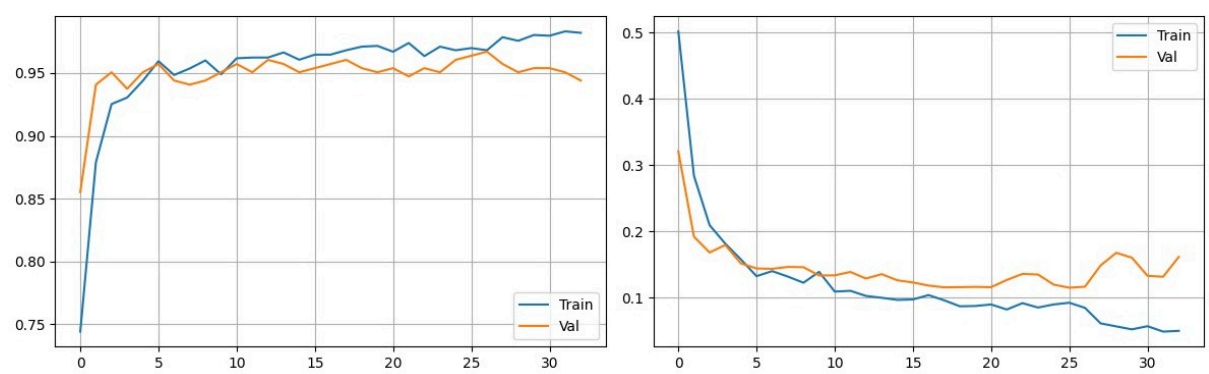


Figure 4: Accuracy & Loss Curves of VGG16



Figure 5: Validation Confusion Matrix of VGG16

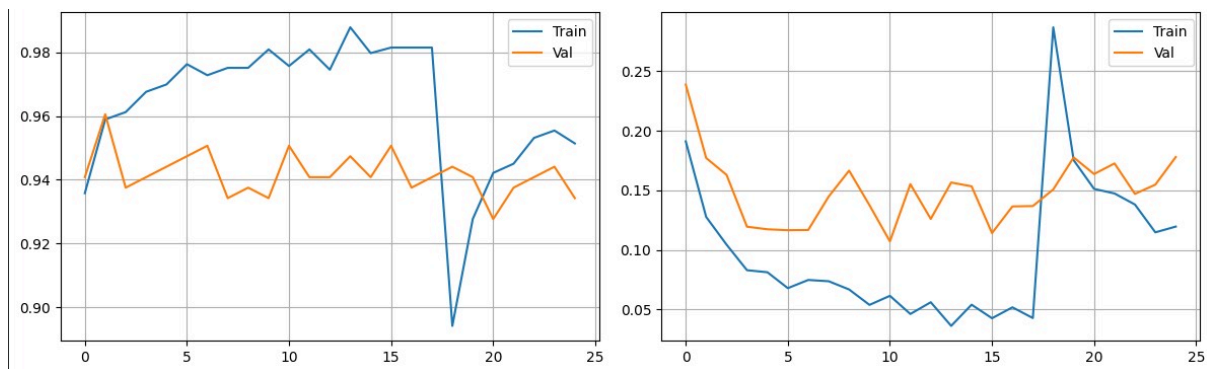


Figure 6: Accuracy & Loss Curves of EfficientNetB0

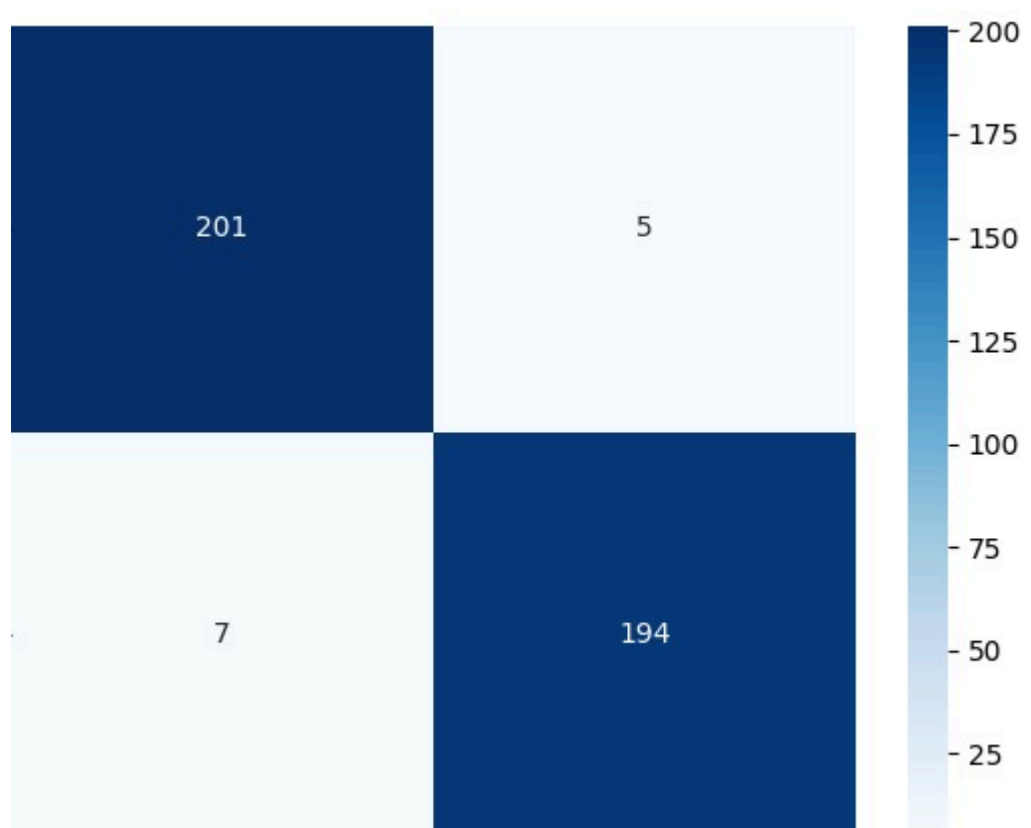


Figure 7: Validation Confusion Matrix of EfficientNetB0

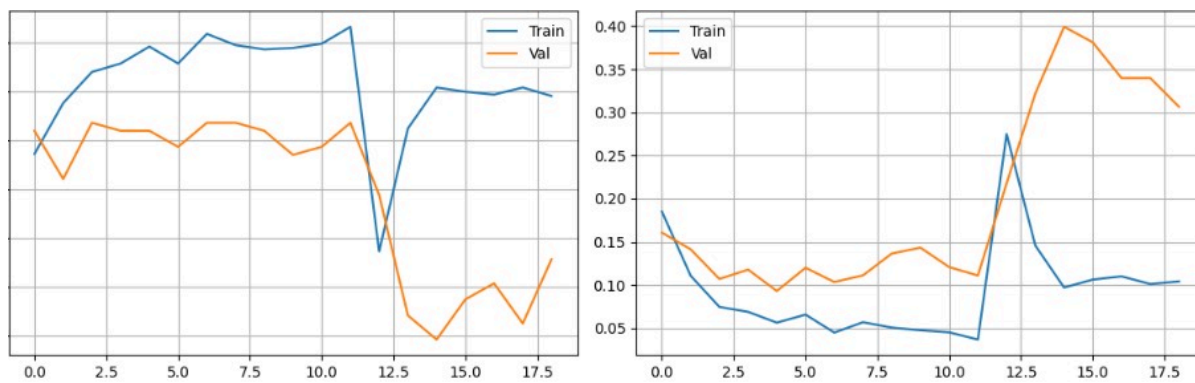


Figure 8: Accuracy & Loss Curves of MobileNetV2



Figure 9: Validation Confusion Matrix of MobileNetV2

5. Discussion

Based on the results presented in Table 2, it can be stated that the proposed ResNet50 model showed the best results compared to others by giving the highest accuracy rate of 97.79%. Furthermore, all of its metrics were balanced, since precision, recall, and F1-score rates were all equal to 0.9779, which reflects high robustness of the suggested model and its efficiency in recognizing CC and Black Spot. VGG16 followed closely with 97.54% accuracy, while EfficientNetB0 reached 97.05% and MobileNetV2 reached 95.33%. The deep residual architecture of ResNet50, combined with transfer learning from ImageNet, allowed the model to capture both fine-grained lesion details and broader contextual features. Moreover, Confusion Matrices, generated during the experiments, confirm the strong performance of the suggested algorithm. Indeed, ResNet50 successfully classified 398 out of 407 images correctly, proving the generated model highly effective in recognizing similar diseases.

While the proposed model achieves strong accuracy, the current study is limited to the binary classification of two specific diseases, focusing only on CC and Black Spot. Additionally, the system currently operates on pre-acquired datasets, and has not yet been integrated into a mobile application for real-time field diagnosis. Future efforts will be required to increase the size of the dataset to include other types of diseases and address these constraints.

6. Conclusion and Future Work

This work succeeded in proposing a reliable and effective DL-based solution for automated detection and classification of citrus disease images into either CC or Black Spot classes using a ResNet50 deep network with transfer learning capability, which yielded a strong accuracy of 97.79%. Comparisons indicated that the proposed algorithm is superior to other algorithms such as MobileNetV2, VGG16, and EfficientNetB0 by having a high balance in performance. Overall, it was found that the proposed approach could be a very helpful and reliable tool in assisting in the disease management process in agriculture without involving manual experts.

Some recommendations for future research based on the current study results are made. First, the dataset is going to be enriched by various other citrus disease images as well as various citrus pest infections. In this way, the diagnostic system will be able to detect a broader range of illnesses.

Next, the goal of future work is the implementation of the trained ResNet50 network into an intuitive user interface in the form of mobile or edge computing applications for farmers to use on site for disease identification. Finally, the trained model should be tested under real-life circumstances without any restrictions.

7. References

[1] Chakraborty, S., Goel, K., Rasal, V., Paul, K., & Mandal, D. (2025). A comprehensive review: exploring bioactive compounds of citrus fruit peels for therapeutic and industrial applications. *Food Science and Engineering*, 54-69.

[2] Algheshairy, R. M., & Ali, H. A. (2026). Antioxidant, anti-inflammatory, and immunomodulatory pathways of Citrus limon juice in a respiratory-irritated rat model. *Frontiers in Nutrition*, 13, 1743575.

[3] Olabinjo, O. O. (2025). Citrus Peels an Effective Sources of Bioactive Compounds. In *Waste Management for a Sustainable Future-Technologies, Strategies and Global Perspectives*. IntechOpen.

[4] Beshara, S., Kassem, A., Fors, H., & Harraz, N. (2024). A Comprehensive Review and Mapping Citrus Supply Chains from a Sustainability Perspective across the European Union, Middle East, and Africa. *Sustainability*, 16(19), 8582.

[5] Johnson, J. B., Hungerford, N. L., Sultanbawa, Y., & Netzel, M. E. (2025). Unlocking the sublime: A review of native Australian citrus species. *Foods*, 14(14), 2425.

[6] Chen, Y., Dai, X., Qiu, A., Liu, W., Li, D., & Li, R. (2026). Genome-wide identification and functional characterization of transmembrane proteins in *Citrus sinensis* during plant-pathogen interaction. *Plant Molecular Biology Reporter*.

[7] Xu, X., Shi, J., Guo, F., Gao, R., Jiang, N., Li, J., & et al. (2026). Sequence typing and whole genome single-nucleotide polymorphisms reveal phylogenetic diversity and population structure of *Xanthomonas citri* subsp. *citri*. *Journal of Applied Microbiology*.

[8] Li, W., He, L., Guan, B., Zeng, X., & et al. (2026). Whole-genome analysis of a historical strain of *Xanthomonas citri* pv. *citri* reveals structural variations. *International Journal of Molecular Sciences*.

[9] Yadav, P. K., Burks, T., Qin, J., Kim, M., Frederick, Q., Dewdney, M. M., & Ritenour, M. A. (2024). Automated classification of citrus disease on fruits and leaves using convolutional neural network generated features from hyperspectral images and machine learning classifiers. *Journal of Applied Remote Sensing*, 18(1).

[10] Batool, A., Safdar, H., & Rehman, F. U. (2026). Application of Image processing and transfer learning for the detection and classification of blemishes and canker diseases of citrus. *Sarhad Journal of Agriculture*, 42(2), 601-608.

[11] Banerjee, S., Mallick, T., Chakraborty, A., Saha, H. N., & Takur, N. T. (2025). Evaluation of State-of-the-Art Deep Learning Techniques for Plant Disease and Pest Detection. *arXiv preprint arXiv:2508.08317*.

- [12] Shreya, S., Rakesh, K. N., Preethi, H. K., Shaan, S., & Divya, H. T. (2026). Intelligent multi-class classification and diagnosis of citrus leaf diseases using deep convolutional network. In E3S Web of Conferences (Vol. 692, p. 03006). EDP Sciences.
- [13] Feng, W., Liu, J., Li, Z., & Lyu, S. (2025). YOLO-Citrus: a lightweight and efficient model for citrus leaf disease detection in complex agricultural environments. *Frontiers in Plant Science*, 16, 1668036.
- [14] Devora-Guadarrama, M., Luna-Benoso, B., Alarcón-Paredes, A., Martínez-Perales, J. C., & Morales-Rodríguez, Ú. S. (2025). Deep learning-based citrus canker and Huanglongbing disease detection using leaf images. *Computers*, 14(11), 500.
- [15] Phung, T. T., Chu, H. D., Nguyen, M. Q., Hoang, T. M., & Pham, M. T. (2025, March). Citrus Disease Detection Using Deep Learning on Edge Devices. In *The International Conference on Intelligent Systems & Networks* (pp. 293-299). Singapore: Springer Nature Singapore..
- [16] Mo, H., & Wei, L. (2024). Lightweight citrus leaf disease detection model based on ARMS and cross-domain dynamic attention. *Journal of King Saud University - Computer and Information Sciences*, 36(7), 102133.
- [17] Ijaz, H. (2024). Citrus disease classification using hybrid deep learning and machine learning models: A case study. *Journal of Agricultural Research*, 62(4), 285-296.
- [18] Kabir, M., Unal, F., Akinci, T. C., Martinez-Morales, A. A., & Ekici, S. (2024). Revealing GLCM metric variations across a plant disease dataset: A comprehensive examination and future prospects for enhanced deep learning applications. *Electronics*, 13(12), 2299.
- [19] Kumar, R., & Singh, V. (2025). Advancing plant disease detection: A comparative analysis of deep learning and hybrid machine learning models. *Machine Graphics & Vision*, 34(3), 57-75.
- [20] Xue, R., & Wang, L. (2025). Research on lightweight citrus leaf pest and disease detection based on PEW-YOLO. *Processes*, 13(5), 1365.
- [21] Ismail, R. N., Hasoon, A. N., Abdulqader, I. R., & Abdulateef, S. K. (2025). ROI - Enhancing detection of citrus disease based on YOLOv10. *Mesopotamian Journal of Computer Science*, 2025, 316-328.
- [22] Rukuna, A. L., Zambuk, F. U., Gital, A. Y., & Bello, U. M. (2025). Citrus diseases detection and classification based on EfficientNet-B5. *Systems and Soft Computing*, 7, 200199.
- [23] Butt, N., Iqbal, M. M., Ramzan, S., Raza, A., Abualigah, L., Fitriyani, N. L., & Syafrudin, M. (2025). Citrus diseases detection using innovative deep learning approach and hybrid meta-heuristic. *PLOS ONE*, 20(1), e0316081.
- [24] Sireesha, N. V., Rekha, G., & Almenweer, R. A. (2025). Unassailable citrus disease classification via multi-stage deep ensemble learning with vision transformers. *Scientific Reports*, 15, 40573.
- [25] Rani, R., & Gupta, S. (2024, August). ResNet50 for accurate plant leaf disease detection and classification. In *2024 Second International Conference on Intelligent Cyber Physical Systems and Internet of Things (ICoICI)* (pp. 1285-1289). IEEE.
- [26] Goyal, P., Gill, J., Goyal, R., Sharma, M., & Sharma, K. (2026). Deep learning-based citrus plant disease classification using a computationally efficient CNN model. *Scientific Reports*.

- [27] Hidayah, N., Muiz, A., & Sunandar, D. (2025). Web-Based Citrus Fruit Disease Detection Application Using MobileNet V2 for Agricultural Quality Assurance. *bit-Tech*, 8(2).
- [28] Chauhan, S. (2024, October). Leveraging MobileNetV2 for accurate detection and classification of orange diseases: a study on blackspot, canker, and greening citrus. In *2024 Global Conference on Communications and Information Technologies (GCCIT)* (pp. 1-6). IEEE.
- [29] Shehada, I. K., Atallah, R. R., & Maghari, A. Y. (2025). Comparative Study of Deep Learning Optimizers: SGD, Momentum SGD, RMSProp, AMSGrad, Adam, Yogi, and Lion.
- [30] Devi, V. S., Pooja, G., Ramya, S., & Priya, N. V. (2025). How deep learning identifies and learns aspects of plant for classification. In *ITM Web of Conferences* (Vol. 74, p. 01013). EDP Sciences.